Modeling Neutron Stars
Lecture-7

Chiranjib Mondal
LPC, Université Caen Normandie
https://sites.google.com/view/nucleartheory/home

UNIVERSITÉ CAEN NORMANDIE

Nuclear Theory Group
February 6, 2021

Contents:

Elementary Field Theory

- Infinite Nuclear matter
- Recap :$\sigma - \omega$ model
- Energy density
- Improvements on $\sigma - \omega$ model

Nuclear Matter

Weizsacker's mass formula

- Saturated system

Nuclear Matter

Weizsacker's mass formula

- Saturated system $\longrightarrow$ short range and strong repulsion at short distance of the force.

Nuclear Matter

Weizsacker's mass formula

- Saturated system $\longrightarrow$ short range and strong repulsion at short distance of the force.
- Weizsacker's mass formula in its simplest form is given by,

$$M(A,Z) = \mathtt{a}_v A + a_{sym}\frac{(N-Z)^2}{A} + a_{surf}A^{2/3} + a_{coul}\frac{Z^2}{A^{1/3}}$$

Nuclear Matter

Weizsacker's mass formula

- Saturated system $\longrightarrow$ short range and strong repulsion at short distance of the force.
- Weizsacker's mass formula in its simplest form is given by,

$$\begin{aligned} M(A,Z) &= \mathtt{a}_v A + a_{sym}\frac{(N-Z)^2}{A} + a_{surf}A^{2/3} + a_{coul}\frac{Z^2}{A^{1/3}} \\ B(A,Z) &= M(A,Z) - \{(A-Z)m_n + Zm_p\} \end{aligned}$$

Nuclear Matter

Weizsacker's mass formula

- Saturated system $\longrightarrow$ short range and strong repulsion at short distance of the force.
- Weizsacker's mass formula in its simplest form is given by,

$$\begin{aligned} M(A,Z) &= \mathtt{a}_v A + a_{sym}\frac{(N-Z)^2}{A} + a_{surf}A^{2/3} + a_{coul}\frac{Z^2}{A^{1/3}} \\ B(A,Z) &= M(A,Z) - \{(A-Z)m_n + Zm_p\} \simeq M(A,Z) - Am \end{aligned}$$

Nuclear Matter

Weizsacker's mass formula

- Saturated system $\longrightarrow$ short range and strong repulsion at short distance of the force.
- Weizsacker's mass formula in its simplest form is given by,

$$\begin{aligned} M(A,Z) &= \mathtt{a}_v A + a_{sym}\frac{(N-Z)^2}{A} + a_{surf}A^{2/3} + a_{coul}\frac{Z^2}{A^{1/3}} \\ B(A,Z) &= M(A,Z) - \{(A-Z)m_n + Zm_p\} \simeq M(A,Z) - Am \\ \Rightarrow \frac{B}{A} &= \frac{M}{A} - m \end{aligned}$$

Nuclear Matter

Weizsacker's mass formula

- Saturated system $\longrightarrow$ short range and strong repulsion at short distance of the force.
- Weizsacker's mass formula in its simplest form is given by,

$$
\begin{aligned}
M(A,Z) &= \mathtt{a}_v A + a_{sym}\frac{(N-Z)^2}{A} + a_{surf}A^{2/3} + a_{coul}\frac{Z^2}{A^{1/3}} \\
B(A,Z) &= M(A,Z) - \{(A-Z)m_n + Zm_p\} \simeq M(A,Z) - Am \\
\Rightarrow \frac{B}{A} &= \frac{M}{A} - m = \mathtt{a}_v + a_{sym}\delta^2 + a_{surf}\frac{1}{A^{1/3}} + a_{coul}\frac{Z^2}{A^{4/3}} - m
\end{aligned}
$$

Nuclear Matter

Weizsacker's mass formula

- Saturated system $\longrightarrow$ short range and strong repulsion at short distance of the force.
- Weizsacker's mass formula in its simplest form is given by,

$$
\begin{aligned}
M(A,Z) &= \mathtt{a}_v A + a_{sym}\frac{(N-Z)^2}{A} + a_{surf}A^{2/3} + a_{coul}\frac{Z^2}{A^{1/3}} \\
B(A,Z) &= M(A,Z) - \{(A-Z)m_n + Zm_p\} \simeq M(A,Z) - Am \\
\Rightarrow \frac{B}{A} &= \frac{M}{A} - m = \mathtt{a}_v + a_{sym}\delta^2 + a_{surf}\frac{1}{A^{1/3}} + a_{coul}\frac{Z^2}{A^{4/3}} - m \\
\Rightarrow \frac{B}{A} &= a_v + a_{sym}\delta^2 + a_{surf}\frac{1}{A^{1/3}} + a_{coul}\frac{Z^2}{A^{4/3}}
\end{aligned}
$$

Nuclear Matter

Weizsacker's mass formula

- Saturated system $\longrightarrow$ short range and strong repulsion at short distance of the force.
- Weizsacker's mass formula in its simplest form is given by,

$$
\begin{aligned}
M(A,Z) &= \mathtt{a}_v A + a_{sym}\frac{(N-Z)^2}{A} + a_{surf}A^{2/3} + a_{coul}\frac{Z^2}{A^{1/3}} \\
B(A,Z) &= M(A,Z) - \{(A-Z)m_n + Zm_p\} \simeq M(A,Z) - Am \\
\Rightarrow \frac{B}{A} &= \frac{M}{A} - m = \mathtt{a}_v + a_{sym}\delta^2 + a_{surf}\frac{1}{A^{1/3}} + a_{coul}\frac{Z^2}{A^{4/3}} - m \\
\Rightarrow \frac{B}{A} &= a_v + a_{sym}\delta^2 + a_{surf}\frac{1}{A^{1/3}} + a_{coul}\frac{Z^2}{A^{4/3}}
\end{aligned}
$$

where, $\delta \rightarrow \left(\frac{N-Z}{A}\right)^2$ is the asymmetry

Nuclear Matter

Weizsacker's mass formula

- Saturated system $\longrightarrow$ short range and strong repulsion at short distance of the force.
- Weizsacker's mass formula in its simplest form is given by,

$$
\begin{aligned}
M(A,Z) &= \mathtt{a}_v A + a_{sym}\frac{(N-Z)^2}{A} + a_{surf}A^{2/3} + a_{coul}\frac{Z^2}{A^{1/3}} \\
B(A,Z) &= M(A,Z) - \{(A-Z)m_n + Zm_p\} \simeq M(A,Z) - Am \\
\Rightarrow \frac{B}{A} &= \frac{M}{A} - m = \mathtt{a}_v + a_{sym}\delta^2 + a_{surf}\frac{1}{A^{1/3}} + a_{coul}\frac{Z^2}{A^{4/3}} - m \\
\Rightarrow \frac{B}{A} &= a_v + a_{sym}\delta^2 + a_{surf}\frac{1}{A^{1/3}} + a_{coul}\frac{Z^2}{A^{4/3}}
\end{aligned}
$$

where, $\delta \rightarrow \left(\frac{N-Z}{A}\right)^2$ is the asymmetry and $a_v = \mathtt{a}_v - m$ is the usual volume term in the formula.

Nuclear Matter

Weizsacker's mass formula

- Saturated system $\longrightarrow$ short range and strong repulsion at short distance of the force.
- Weizsacker's mass formula in its simplest form is given by,

$$\begin{aligned} M(A,Z) &= \mathtt{a}_v A + a_{sym}\frac{(N-Z)^2}{A} + a_{surf}A^{2/3} + a_{coul}\frac{Z^2}{A^{1/3}} \\ B(A,Z) &= M(A,Z) - \{(A-Z)m_n + Zm_p\} \simeq M(A,Z) - Am \\ \Rightarrow \frac{B}{A} &= \frac{M}{A} - m = \mathtt{a}_v + a_{sym}\delta^2 + a_{surf}\frac{1}{A^{1/3}} + a_{coul}\frac{Z^2}{A^{4/3}} - m \\ \Rightarrow \frac{B}{A} &= a_v + a_{sym}\delta^2 + a_{surf}\frac{1}{A^{1/3}} + a_{coul}\frac{Z^2}{A^{4/3}} \end{aligned}$$

where, $\delta \to \left(\frac{N-Z}{A}\right)^2$ is the asymmetry and $a_v = \mathtt{a}_v - m$ is the usual volume term in the formula.

- From Binding energy data we have

$$a_v \simeq -16 \text{ MeV}, \qquad a_{sym} \simeq 32 \text{ MeV}.$$

Nuclear Matter

Weizsacker's mass formula

- Saturated system $\longrightarrow$ short range and strong repulsion at short distance of the force.
- Weizsacker's mass formula in its simplest form is given by,

$$\begin{aligned} M(A,Z) &= \mathtt{a}_v A + a_{sym}\frac{(N-Z)^2}{A} + a_{surf}A^{2/3} + a_{coul}\frac{Z^2}{A^{1/3}} \\ B(A,Z) &= M(A,Z) - \{(A-Z)m_n + Zm_p\} \simeq M(A,Z) - Am \\ \Rightarrow \frac{B}{A} &= \frac{M}{A} - m = \mathtt{a}_v + a_{sym}\delta^2 + a_{surf}\frac{1}{A^{1/3}} + a_{coul}\frac{Z^2}{A^{4/3}} - m \\ \Rightarrow \frac{B}{A} &= a_v + a_{sym}\delta^2 + a_{surf}\frac{1}{A^{1/3}} + a_{coul}\frac{Z^2}{A^{4/3}} \end{aligned}$$

  where, $\delta \to \left(\frac{N-Z}{A}\right)^2$ is the asymmetry and $a_v = \mathtt{a}_v - m$ is the usual volume term in the formula.
- From Binding energy data we have

$$a_v \simeq -16 \text{ MeV}, \qquad a_{sym} \simeq 32 \text{ MeV}.$$

- For symmetric infinite matter $\frac{B}{A} \simeq -16$ MeV.

Nuclear Matter

Weizsacker's mass formula

- Radius of a nucleus is parametrized by $R = r_0 A^{1/3}$

Nuclear Matter

Weizsacker's mass formula

- Radius of a nucleus is parametrized by $R = r_0 A^{1/3} \Rightarrow r_0 \sim 1.15$ fm

Nuclear Matter

Weizsacker's mass formula

- Radius of a nucleus is parametrized by $R = r_0 A^{1/3} \Rightarrow r_0 \sim 1.15$ fm $\Rightarrow \rho_0 = \left[(4\pi/3)\, r_0^3\right]^{-1}$

Nuclear Matter

Weizsacker's mass formula

- Radius of a nucleus is parametrized by $R = r_0 A^{1/3} \Rightarrow r_0 \sim 1.15$ fm $\Rightarrow \rho_0 = \left[(4\pi/3)\, r_0^3\right]^{-1} \simeq 0.16$ fm^{-3}.

Nuclear Matter

Weizsacker's mass formula

- Radius of a nucleus is parametrized by $R = r_0 A^{1/3} \Rightarrow r_0 \sim 1.15$ fm $\Rightarrow \rho_0 = \left[(4\pi/3)\, r_0^3\right]^{-1} \simeq 0.16$ fm^{-3}.
- Let's associate ϵ_0 as the saturation energy density

Nuclear Matter

Weizsacker's mass formula

- Radius of a nucleus is parametrized by $R = r_0 A^{1/3} \Rightarrow r_0 \sim 1.15$ fm $\Rightarrow \rho_0 = \left[(4\pi/3)\, r_0^3\right]^{-1} \simeq 0.16$ fm^{-3}.
- Let's associate ϵ_0 as the saturation energy density $\Rightarrow \mathtt{a}_v = (4\pi/3)\, r_0^3 \epsilon_0$.

Nuclear Matter

Weizsacker's mass formula

- Radius of a nucleus is parametrized by $R = r_0 A^{1/3} \Rightarrow r_0 \sim 1.15$ fm $\Rightarrow \rho_0 = \left[(4\pi/3)\, r_0^3\right]^{-1} \simeq 0.16$ fm^{-3}.
- Let's associate ϵ_0 as the saturation energy density $\Rightarrow \mathtt{a}_v = (4\pi/3)\, r_0^3 \epsilon_0$.
- If we concentrate on infinite symmetric matter without Coulomb,

$$\frac{B}{A} = (4\pi/3)\, r_0^3 \epsilon_0 - m$$

Nuclear Matter

Weizsacker's mass formula

- Radius of a nucleus is parametrized by $R = r_0 A^{1/3} \Rightarrow r_0 \sim 1.15$ fm $\Rightarrow \rho_0 = \left[(4\pi/3)\, r_0^3\right]^{-1} \simeq 0.16$ fm^{-3}.
- Let's associate ϵ_0 as the saturation energy density $\Rightarrow \mathtt{a}_v = (4\pi/3)\, r_0^3 \epsilon_0$.
- If we concentrate on infinite symmetric matter without Coulomb,

$$\frac{B}{A} = (4\pi/3)\, r_0^3 \epsilon_0 - m = \left(\frac{\epsilon}{\rho}\right)_0 - m$$

Nuclear Matter

Weizsacker's mass formula

- Radius of a nucleus is parametrized by $R = r_0 A^{1/3} \Rightarrow r_0 \sim 1.15$ fm $\Rightarrow \rho_0 = \left[(4\pi/3)\, r_0^3\right]^{-1} \simeq 0.16$ fm^{-3}.
- Let's associate ϵ_0 as the saturation energy density $\Rightarrow \mathtt{a}_v = (4\pi/3)\, r_0^3 \epsilon_0$.
- If we concentrate on infinite symmetric matter without Coulomb,

$$\frac{B}{A} = (4\pi/3)\, r_0^3 \epsilon_0 - m = \left(\frac{\epsilon}{\rho}\right)_0 - m \Rightarrow \epsilon_0 \sim 147 \text{ MeV fm}^{-3}.$$

Nuclear Matter

Microscopic description

- Energy per particle of nuclear matter of density $\rho = \rho_{\rm n} + \rho_{\rm p}$ and asymmetry $\delta = (\rho_{\rm n} - \rho_{\rm p})/\rho$ is given by,

$$\frac{\epsilon}{\rho}(\rho, \delta) = \frac{\epsilon}{\rho}(\rho, 0) + e_{\rm sym}(\rho)\delta^2 + \mathcal{O}(\delta^4),$$

Nuclear Matter

Microscopic description

- Energy per particle of nuclear matter of density $\rho = \rho_{\rm n} + \rho_{\rm p}$ and asymmetry $\delta = (\rho_{\rm n} - \rho_{\rm p})/\rho$ is given by,

$$\frac{\epsilon}{\rho}(\rho, \delta) \quad = \quad \frac{\epsilon}{\rho}(\rho, 0) + e_{\rm sym}(\rho)\delta^2 + \mathcal{O}(\delta^4), \text{ with, } e_{\rm sym}(\rho) = \frac{1}{2}\frac{\partial^2 \left(\frac{\epsilon}{\rho}\right)(\rho, \delta)}{\partial \delta^2}$$

Nuclear Matter

Microscopic description

- Energy per particle of nuclear matter of density $\rho = \rho_{\rm n} + \rho_{\rm p}$ and asymmetry $\delta = (\rho_{\rm n} - \rho_{\rm p})/\rho$ is given by,

$$\begin{aligned}
\frac{\epsilon}{\rho}(\rho,\delta) &= \frac{\epsilon}{\rho}(\rho,0) + e_{\rm sym}(\rho)\delta^2 + \mathcal{O}(\delta^4), \text{ with, } e_{\rm sym}(\rho) = \frac{1}{2}\frac{\partial^2\left(\frac{\epsilon}{\rho}\right)(\rho,\delta)}{\partial\delta^2} \\
\frac{\epsilon}{\rho}(\rho,0) &= \left(\frac{\epsilon}{\rho}\right)_0 + \frac{1}{2}K_0\left(\frac{\rho-\rho_0}{3\rho_0}\right)^2 + \mathcal{O}(\rho^3) + \cdots
\end{aligned}$$

Nuclear Matter

Microscopic description

- Energy per particle of nuclear matter of density $\rho = \rho_{\rm n} + \rho_{\rm p}$ and asymmetry $\delta = (\rho_{\rm n} - \rho_{\rm p})/\rho$ is given by,

$$\begin{aligned}
\frac{\epsilon}{\rho}(\rho,\delta) &= \frac{\epsilon}{\rho}(\rho,0) + e_{\rm sym}(\rho)\delta^2 + \mathcal{O}(\delta^4), \text{ with, } e_{\rm sym}(\rho) = \frac{1}{2}\frac{\partial^2\left(\frac{\epsilon}{\rho}\right)(\rho,\delta)}{\partial\delta^2} \\
\frac{\epsilon}{\rho}(\rho,0) &= \left(\frac{\epsilon}{\rho}\right)_0 + \frac{1}{2}K_0\left(\frac{\rho-\rho_0}{3\rho_0}\right)^2 + \mathcal{O}(\rho^3) + \cdots
\end{aligned}$$

- Here, $K_0 = 9\rho_0^2\left(\frac{\partial^2\frac{\epsilon}{\rho}(\rho,0)}{\partial\rho^2}\right)_{\rho_0}$.

Nuclear Matter

Microscopic description

- Energy per particle of nuclear matter of density $\rho = \rho_{\rm n} + \rho_{\rm p}$ and asymmetry $\delta = (\rho_{\rm n} - \rho_{\rm p})/\rho$ is given by,

$$\begin{aligned}
\frac{\epsilon}{\rho}(\rho,\delta) &= \frac{\epsilon}{\rho}(\rho,0) + e_{\rm sym}(\rho)\delta^2 + \mathcal{O}(\delta^4), \text{ with, } e_{\rm sym}(\rho) = \frac{1}{2}\frac{\partial^2\left(\frac{\epsilon}{\rho}\right)(\rho,\delta)}{\partial\delta^2} \\
\frac{\epsilon}{\rho}(\rho,0) &= \left(\frac{\epsilon}{\rho}\right)_0 + \frac{1}{2}K_0\left(\frac{\rho-\rho_0}{3\rho_0}\right)^2 + \mathcal{O}(\rho^3) + \cdots
\end{aligned}$$

- Here, $K_0 = 9\rho_0^2\left(\frac{\partial^2\frac{\epsilon}{\rho}(\rho,0)}{\partial\rho^2}\right)_{\rho_0}$.
- The symmetry energy is expanded as

$$e_{\rm sym}(\rho) = J + L\left(\frac{\rho-\rho_0}{3\rho_0}\right) + \frac{1}{2}K_{\rm sym}\left(\frac{\rho-\rho_0}{3\rho_0}\right)^2 + \mathcal{O}(\rho^3) + \cdots$$

Nuclear Matter

Microscopic description

- Energy per particle of nuclear matter of density $\rho = \rho_{\rm n} + \rho_{\rm p}$ and asymmetry $\delta = (\rho_{\rm n} - \rho_{\rm p})/\rho$ is given by,

$$\begin{aligned}\frac{\epsilon}{\rho}(\rho,\delta) &= \frac{\epsilon}{\rho}(\rho,0) + e_{\rm sym}(\rho)\delta^2 + \mathcal{O}(\delta^4), \text{ with, } e_{\rm sym}(\rho) = \frac{1}{2}\frac{\partial^2\left(\frac{\epsilon}{\rho}\right)(\rho,\delta)}{\partial\delta^2} \\ \frac{\epsilon}{\rho}(\rho,0) &= \left(\frac{\epsilon}{\rho}\right)_0 + \frac{1}{2}K_0\left(\frac{\rho-\rho_0}{3\rho_0}\right)^2 + \mathcal{O}(\rho^3) + \cdots\end{aligned}$$

- Here, $K_0 = 9\rho_0^2\left(\frac{\partial^2\frac{\epsilon}{\rho}(\rho,0)}{\partial\rho^2}\right)_{\rho_0}$.
- The symmetry energy is expanded as

$$e_{\rm sym}(\rho) = J + L\left(\frac{\rho-\rho_0}{3\rho_0}\right) + \frac{1}{2}K_{\rm sym}\left(\frac{\rho-\rho_0}{3\rho_0}\right)^2 + \mathcal{O}(\rho^3) + \cdots$$

- Here, $J = e_{\rm sym}(\rho_0)$, $L = \left(3\rho\frac{\partial e_{\rm sym}}{\partial\rho}\right)_{\rho_0}$, $K_{\rm sym} = 9\rho_0^2\left(\frac{\partial^2 e_{\rm sym}}{\partial\rho^2}\right)_{\rho_0}$.

Nuclear Matter

Microscopic description

- Energy per particle of nuclear matter of density $\rho = \rho_{\rm n} + \rho_{\rm p}$ and asymmetry $\delta = (\rho_{\rm n} - \rho_{\rm p})/\rho$ is given by,

$$\begin{aligned}\frac{\epsilon}{\rho}(\rho,\delta) &= \frac{\epsilon}{\rho}(\rho,0) + e_{\rm sym}(\rho)\delta^2 + \mathcal{O}(\delta^4), \text{ with, } e_{\rm sym}(\rho) = \frac{1}{2}\frac{\partial^2\left(\frac{\epsilon}{\rho}\right)(\rho,\delta)}{\partial\delta^2} \\ \frac{\epsilon}{\rho}(\rho,0) &= \left(\frac{\epsilon}{\rho}\right)_0 + \frac{1}{2}K_0\left(\frac{\rho-\rho_0}{3\rho_0}\right)^2 + \mathcal{O}(\rho^3) + \cdots\end{aligned}$$

- Here, $K_0 = 9\rho_0^2\left(\frac{\partial^2\frac{\epsilon}{\rho}(\rho,0)}{\partial\rho^2}\right)_{\rho_0}$.
- The symmetry energy is expanded as

$$e_{\rm sym}(\rho) = J + L\left(\frac{\rho-\rho_0}{3\rho_0}\right) + \frac{1}{2}K_{\rm sym}\left(\frac{\rho-\rho_0}{3\rho_0}\right)^2 + \mathcal{O}(\rho^3) + \cdots$$

- Here, $J = e_{\rm sym}(\rho_0)$, $L = \left(3\rho\frac{\partial e_{\rm sym}}{\partial\rho}\right)_{\rho_0}$, $K_{\rm sym} = 9\rho_0^2\left(\frac{\partial^2 e_{\rm sym}}{\partial\rho^2}\right)_{\rho_0}$.
- The pressure is given by, $\mathtt{p}(\rho) = \rho^2\frac{\partial\left(\frac{\epsilon}{\rho}\right)(\rho,0)}{\partial\rho}$.

$\sigma - \omega$ model

Under approximations

$\sigma - \omega$ model

Under approximations

- The "*stress-energy tensor*" is defined as (Lec-3)

$$\boxed{T^{\mu\nu} = \frac{\partial \mathscr{L}}{\partial(\partial_\mu \Phi^{\mathrm{A}})}\partial^\nu \Phi^{\mathrm{A}} - g^{\mu\nu}\mathscr{L}}.$$

$\sigma - \omega$ model

Under approximations

- The "*stress-energy tensor*" is defined as (Lec-3)

$$\boxed{T^{\mu\nu} = \frac{\partial \mathscr{L}}{\partial(\partial_\mu \Phi^{\mathrm{A}})} \partial^\nu \Phi^{\mathrm{A}} - g^{\mu\nu} \mathscr{L}}.$$

We saw that $T^{00} = \mathscr{H}$ and $T^{ii} = \mathtt{p}$.

$\sigma - \omega$ model

Under approximations

- The "*stress-energy tensor*" is defined as (Lec-3)

$$\boxed{T^{\mu\nu} = \frac{\partial \mathscr{L}}{\partial(\partial_\mu \Phi^{\mathrm{A}})}\partial^\nu \Phi^{\mathrm{A}} - g^{\mu\nu}\mathscr{L}}.$$

  We saw that $T^{00} = \mathscr{H}$ and $T^{ii} = \mathtt{p}$.

- The Lagrangian of $\sigma - \omega$ model is given by,

$$\begin{aligned}\mathscr{L} &= \overline{\psi}\left[i\gamma^\mu\left(\partial_\mu + i\mathrm{g}_\omega \omega_\mu\right) - \left(m - \mathrm{g}_\sigma \sigma\right)\right]\psi \\ &\quad + \frac{1}{2}\left(\partial_\mu \sigma \partial^\mu \sigma - m_\sigma^2 \sigma^2\right) - \frac{1}{4}\Omega_{\mu\nu}\Omega^{\mu\nu} + \frac{1}{2}m_\omega^2 \omega_\mu \omega^\mu.\end{aligned}$$

$\sigma - \omega$ model

Under approximations

- The "*stress-energy tensor*" is defined as (Lec-3)

$$\boxed{T^{\mu\nu} = \frac{\partial \mathscr{L}}{\partial(\partial_\mu \Phi^{\mathrm{A}})}\partial^\nu \Phi^{\mathrm{A}} - g^{\mu\nu}\mathscr{L}}.$$

  We saw that $T^{00} = \mathscr{H}$ and $T^{ii} = \mathtt{p}$.

- The Lagrangian of $\sigma - \omega$ model is given by,

$$\begin{aligned}\mathscr{L} &= \overline{\psi}\left[i\gamma^\mu\left(\partial_\mu + i\mathrm{g}_\omega \omega_\mu\right) - \left(m - \mathrm{g}_\sigma \sigma\right)\right]\psi \\ &\quad + \frac{1}{2}\left(\partial_\mu \sigma \partial^\mu \sigma - m_\sigma^2 \sigma^2\right) - \frac{1}{4}\Omega_{\mu\nu}\Omega^{\mu\nu} + \frac{1}{2}m_\omega^2 \omega_\mu \omega^\mu.\end{aligned}$$

- Under transformation
  $\psi \longrightarrow e^{-iq\theta}\psi$

$\sigma - \omega$ model

Under approximations

- The "*stress-energy tensor*" is defined as (Lec-3)

$$\boxed{T^{\mu\nu} = \frac{\partial \mathscr{L}}{\partial(\partial_\mu \Phi^{\mathrm{A}})}\partial^\nu \Phi^{\mathrm{A}} - g^{\mu\nu}\mathscr{L}}.$$

  We saw that $T^{00} = \mathscr{H}$ and $T^{ii} = \mathtt{p}$.

- The Lagrangian of $\sigma - \omega$ model is given by,

$$\begin{aligned}\mathscr{L} &= \overline{\psi}\left[i\gamma^\mu\left(\partial_\mu + i\mathrm{g}_\omega \omega_\mu\right) - \left(m - \mathrm{g}_\sigma \sigma\right)\right]\psi \\ &\quad + \frac{1}{2}\left(\partial_\mu \sigma \partial^\mu \sigma - m_\sigma^2 \sigma^2\right) - \frac{1}{4}\Omega_{\mu\nu}\Omega^{\mu\nu} + \frac{1}{2}m_\omega^2 \omega_\mu \omega^\mu.\end{aligned}$$

- Under transformation
  $\psi \longrightarrow e^{-iq\theta}\psi \Rightarrow j^\mu = q\overline{\psi}\gamma^\mu\psi$

$\sigma - \omega$ model

Under approximations

- The "*stress-energy tensor*" is defined as (Lec-3)

$$\boxed{T^{\mu\nu} = \frac{\partial \mathscr{L}}{\partial(\partial_\mu \Phi^{\mathrm{A}})}\partial^\nu \Phi^{\mathrm{A}} - g^{\mu\nu}\mathscr{L}}.$$

  We saw that $T^{00} = \mathscr{H}$ and $T^{ii} = \mathtt{p}$.

- The Lagrangian of $\sigma - \omega$ model is given by,

$$\begin{aligned}\mathscr{L} &= \overline{\psi}\left[i\gamma^\mu\left(\partial_\mu + i\mathrm{g}_\omega \omega_\mu\right) - \left(m - \mathrm{g}_\sigma \sigma\right)\right]\psi \\ &\quad + \frac{1}{2}\left(\partial_\mu \sigma \partial^\mu \sigma - m_\sigma^2 \sigma^2\right) - \frac{1}{4}\Omega_{\mu\nu}\Omega^{\mu\nu} + \frac{1}{2}m_\omega^2 \omega_\mu \omega^\mu.\end{aligned}$$

- Under transformation
  $\psi \longrightarrow e^{-iq\theta}\psi \Rightarrow j^\mu = q\overline{\psi}\gamma^\mu\psi \Rightarrow \partial_\mu j^\mu = \partial_\mu \omega^\mu = 0.$

$\sigma - \omega$ model

Under approximations

- Under time reversal $\psi \to \widetilde{\psi} = T\psi^* = i\gamma^1\gamma^3\psi^*$

$\sigma - \omega$ model

Under approximations

- Under time reversal $\psi \to \widetilde{\psi} = T\psi^* = i\gamma^1\gamma^3\psi^* \Rightarrow j^i = 0.$

$\sigma - \omega$ model

Under approximations

- Under time reversal $\psi \to \widetilde{\psi} = T\psi^* = i\gamma^1\gamma^3\psi^* \Rightarrow j^i = 0$.
- Field equations under these symmetries,

$$(\Box + m_\sigma^2)\sigma = g_\sigma \overline{\psi}\psi = g_\sigma \psi^\dagger \gamma^0 \psi.$$
$$(\Box + m_\omega^2)\omega^0 = g_\omega j^0 = g_\omega \psi^\dagger \psi.$$
$$\left[\gamma^\mu \left(i\partial_\mu - g_\omega \omega_\mu\right) - \left(m - g_\sigma \sigma\right)\right]\psi(x) = 0.$$

$\sigma - \omega$ model

Under approximations

- Under time reversal $\psi \to \widetilde{\psi} = T\psi^* = i\gamma^1\gamma^3\psi^* \Rightarrow j^i = 0$.
- Field equations under these symmetries,

$$(\Box + m_\sigma^2)\sigma = g_\sigma \overline{\psi}\psi = g_\sigma \psi^\dagger \gamma^0 \psi.$$
$$(\Box + m_\omega^2)\omega^0 = g_\omega j^0 = g_\omega \psi^\dagger \psi.$$
$$\left[\gamma^\mu \left(i\partial_\mu - g_\omega \omega_\mu\right) - \left(m - g_\sigma \sigma\right)\right]\psi(x) = 0.$$

- Field equations under static and mean-field approximation,

$$\boxed{g_\sigma \langle \sigma \rangle = \left(\frac{g_\sigma}{m_\sigma}\right)^2 \langle \overline{\psi}\psi \rangle = \left(\frac{g_\sigma}{m_\sigma}\right)^2 \frac{2}{\pi^2} \int_0^{k_f} p^2\, dp \frac{m - g_\sigma \sigma}{\sqrt{p^2 + (m - g_\sigma \sigma)^2}}},$$

$$\boxed{g_\omega \langle \omega \rangle = \left(\frac{g_\omega}{m_\omega}\right)^2 \langle \psi^\dagger \psi \rangle = \left(\frac{g_\omega}{m_\omega}\right)^2 \rho_b}.$$

Stress-Energy Tensor

Energy of the σ, ω fields

Stress-Energy Tensor

Energy of the σ, ω fields

- *Stress-Energy tensor* for a field Φ^{A} is defined as

$$T^{\mu\nu} = \frac{\partial \mathscr{L}}{\partial(\partial_\mu \Phi^{\mathrm{A}})} \partial^\nu \Phi^{\mathrm{A}} - g^{\mu\nu} \mathscr{L},$$

Stress-Energy Tensor

Energy of the σ, ω fields

- *Stress-Energy tensor* for a field Φ^{A} is defined as

$$T^{\mu\nu} = \frac{\partial \mathscr{L}}{\partial(\partial_\mu \Phi^{\mathrm{A}})} \partial^\nu \Phi^{\mathrm{A}} - g^{\mu\nu} \mathscr{L},$$

$$\Rightarrow \quad \mathscr{H} = T^{00} = \frac{\partial \mathscr{L}}{\partial(\partial_0 \Phi^{\mathrm{A}})} \partial^0 \Phi^{\mathrm{A}} - \mathscr{L}.$$

Stress-Energy Tensor

Energy of the σ, ω fields

- *Stress-Energy tensor* for a field Φ^{A} is defined as

$$T^{\mu\nu} = \frac{\partial \mathscr{L}}{\partial(\partial_\mu \Phi^{\mathrm{A}})} \partial^\nu \Phi^{\mathrm{A}} - g^{\mu\nu} \mathscr{L},$$

$$\Rightarrow \quad \mathscr{H} = T^{00} = \frac{\partial \mathscr{L}}{\partial(\partial_0 \Phi^{\mathrm{A}})} \partial^0 \Phi^{\mathrm{A}} - \mathscr{L}.$$

$$\Rightarrow \quad \mathscr{H}_{\sigma,\omega} = -\mathscr{L}_{\sigma,\omega}, \text{ (static field);}$$

Stress-Energy Tensor

Energy of the σ, ω fields

- *Stress-Energy tensor* for a field Φ^{A} is defined as

$$T^{\mu\nu} = \frac{\partial \mathscr{L}}{\partial(\partial_\mu \Phi^{\mathrm{A}})} \partial^\nu \Phi^{\mathrm{A}} - g^{\mu\nu} \mathscr{L},$$

$$\Rightarrow \quad \mathscr{H} = T^{00} = \frac{\partial \mathscr{L}}{\partial(\partial_0 \Phi^{\mathrm{A}})} \partial^0 \Phi^{\mathrm{A}} - \mathscr{L}.$$

$$\Rightarrow \quad \mathscr{H}_{\sigma,\omega} = -\mathscr{L}_{\sigma,\omega}, \text{ (static field);}$$

$$\Rightarrow \quad \mathscr{H}_\sigma = -\frac{1}{2}\left[(\partial_\mu \sigma)(\partial^\mu \sigma) - m_\sigma^2 \sigma^2\right]$$

Stress-Energy Tensor

Energy of the σ, ω fields

- *Stress-Energy tensor* for a field Φ^{A} is defined as

$$T^{\mu\nu} = \frac{\partial \mathscr{L}}{\partial(\partial_\mu \Phi^{\mathrm{A}})} \partial^\nu \Phi^{\mathrm{A}} - g^{\mu\nu} \mathscr{L},$$

$$\Rightarrow \quad \mathscr{H} = T^{00} = \frac{\partial \mathscr{L}}{\partial(\partial_0 \Phi^{\mathrm{A}})} \partial^0 \Phi^{\mathrm{A}} - \mathscr{L}.$$

$$\Rightarrow \quad \mathscr{H}_{\sigma,\omega} = -\mathscr{L}_{\sigma,\omega}, \text{ (static field);}$$

$$\Rightarrow \quad \mathscr{H}_\sigma = -\frac{1}{2}\left[(\partial_\mu \sigma)(\partial^\mu \sigma) - m_\sigma^2 \sigma^2\right] = \frac{1}{2} m_\sigma^2 \sigma^2, \text{ (static mean field);}$$

Stress-Energy Tensor

Energy of the σ, ω fields

- *Stress-Energy tensor* for a field Φ^{A} is defined as

$$T^{\mu\nu} = \frac{\partial \mathscr{L}}{\partial(\partial_\mu \Phi^{\mathrm{A}})} \partial^\nu \Phi^{\mathrm{A}} - g^{\mu\nu} \mathscr{L},$$

$$\Rightarrow \quad \mathscr{H} = T^{00} = \frac{\partial \mathscr{L}}{\partial(\partial_0 \Phi^{\mathrm{A}})} \partial^0 \Phi^{\mathrm{A}} - \mathscr{L}.$$

$$\Rightarrow \quad \mathscr{H}_{\sigma,\omega} = -\mathscr{L}_{\sigma,\omega}, \text{ (static field);}$$

$$\Rightarrow \quad \mathscr{H}_\sigma = -\frac{1}{2}\left[(\partial_\mu \sigma)(\partial^\mu \sigma) - m_\sigma^2 \sigma^2\right] = \frac{1}{2} m_\sigma^2 \sigma^2, \text{ (static mean field);}$$

$$\Rightarrow \quad \mathscr{H}_\omega = \frac{1}{4}\Omega_{\mu\nu}\Omega^{\mu\nu} - \frac{1}{2} m_\omega^2 \omega_\mu \omega^\mu$$

Stress-Energy Tensor

Energy of the σ, ω fields

- *Stress-Energy tensor* for a field Φ^{A} is defined as

$$T^{\mu\nu} = \frac{\partial \mathscr{L}}{\partial(\partial_\mu \Phi^{\mathrm{A}})} \partial^\nu \Phi^{\mathrm{A}} - g^{\mu\nu} \mathscr{L},$$

$$\Rightarrow \quad \mathscr{H} = T^{00} = \frac{\partial \mathscr{L}}{\partial(\partial_0 \Phi^{\mathrm{A}})} \partial^0 \Phi^{\mathrm{A}} - \mathscr{L}.$$

$$\Rightarrow \quad \mathscr{H}_{\sigma,\omega} = -\mathscr{L}_{\sigma,\omega}, \text{ (static field);}$$

$$\Rightarrow \quad \mathscr{H}_\sigma = -\frac{1}{2}\left[(\partial_\mu \sigma)(\partial^\mu \sigma) - m_\sigma^2 \sigma^2\right] = \frac{1}{2} m_\sigma^2 \sigma^2, \text{ (static mean field);}$$

$$\Rightarrow \quad \mathscr{H}_\omega = \frac{1}{4}\Omega_{\mu\nu}\Omega^{\mu\nu} - \frac{1}{2} m_\omega^2 \omega_\mu \omega^\mu$$

$$= \frac{1}{2}(\partial_\mu \omega_\nu \partial^\mu \omega^\nu - \partial_\mu \omega_\nu \partial^\nu \omega^\mu) - \frac{1}{2} m_\omega^2 (\omega_0)^2, \text{ } (\omega^i = 0 \text{ under } T\text{-reversal});$$

Stress-Energy Tensor

Energy of the σ, ω fields

- *Stress-Energy tensor* for a field Φ^{A} is defined as

$$T^{\mu\nu} = \frac{\partial \mathscr{L}}{\partial(\partial_\mu \Phi^{\mathrm{A}})}\partial^\nu \Phi^{\mathrm{A}} - g^{\mu\nu}\mathscr{L},$$

$$\Rightarrow \quad \mathscr{H} = T^{00} = \frac{\partial \mathscr{L}}{\partial(\partial_0 \Phi^{\mathrm{A}})}\partial^0 \Phi^{\mathrm{A}} - \mathscr{L}.$$

$$\Rightarrow \quad \mathscr{H}_{\sigma,\omega} = -\mathscr{L}_{\sigma,\omega}, \text{ (static field);}$$

$$\Rightarrow \quad \mathscr{H}_\sigma = -\frac{1}{2}\left[(\partial_\mu\sigma)(\partial^\mu\sigma) - m_\sigma^2\sigma^2\right] = \frac{1}{2}m_\sigma^2\sigma^2, \text{ (static mean field);}$$

$$\Rightarrow \quad \mathscr{H}_\omega = \frac{1}{4}\Omega_{\mu\nu}\Omega^{\mu\nu} - \frac{1}{2}m_\omega^2\omega_\mu\omega^\mu$$

$$= \frac{1}{2}(\partial_\mu\omega_\nu\partial^\mu\omega^\nu - \partial_\mu\omega_\nu\partial^\nu\omega^\mu) - \frac{1}{2}m_\omega^2(\omega_0)^2, \text{ ($\omega^i = 0$ under T-reversal);}$$

$$= \frac{1}{2}(\partial_i\omega_\nu\partial^i\omega^\nu - \partial_i\omega_\nu\partial^\nu\omega^i) - \frac{1}{2}m_\omega^2(\omega_0)^2, \text{ (static field);}$$

Stress-Energy Tensor

Energy of the σ, ω fields

- *Stress-Energy tensor* for a field Φ^{A} is defined as

$$T^{\mu\nu} = \frac{\partial \mathscr{L}}{\partial(\partial_\mu \Phi^{\mathrm{A}})} \partial^\nu \Phi^{\mathrm{A}} - g^{\mu\nu} \mathscr{L},$$

$$\Rightarrow \quad \mathscr{H} = T^{00} = \frac{\partial \mathscr{L}}{\partial(\partial_0 \Phi^{\mathrm{A}})} \partial^0 \Phi^{\mathrm{A}} - \mathscr{L}.$$

$$\Rightarrow \quad \mathscr{H}_{\sigma,\omega} = -\mathscr{L}_{\sigma,\omega}, \text{ (static field);}$$

$$\Rightarrow \quad \mathscr{H}_\sigma = -\frac{1}{2}\left[(\partial_\mu \sigma)(\partial^\mu \sigma) - m_\sigma^2 \sigma^2\right] = \frac{1}{2} m_\sigma^2 \sigma^2, \text{ (static mean field);}$$

$$\Rightarrow \quad \mathscr{H}_\omega = \frac{1}{4}\Omega_{\mu\nu}\Omega^{\mu\nu} - \frac{1}{2} m_\omega^2 \omega_\mu \omega^\mu$$

$$= \frac{1}{2}(\partial_\mu \omega_\nu \partial^\mu \omega^\nu - \partial_\mu \omega_\nu \partial^\nu \omega^\mu) - \frac{1}{2} m_\omega^2 (\omega_0)^2, \text{ ($\omega^i = 0$ under T-reversal);}$$

$$= \frac{1}{2}(\partial_i \omega_\nu \partial^i \omega^\nu - \partial_i \omega_\nu \partial^\nu \omega^i) - \frac{1}{2} m_\omega^2 (\omega_0)^2, \text{ (static field);}$$

$$= -\frac{1}{2} m_\omega^2 \omega_0^2, \text{ (mean field, $\omega^i = 0$).}$$

Stress-Energy Tensor

Energy of the ψ field in interaction

Stress-Energy Tensor

Energy of the ψ field in interaction

- *Stress-Energy tensor* is defined as

$$T^{\mu\nu} = \frac{\partial \mathscr{L}}{\partial(\partial_\mu \Phi^{\mathrm{A}})} \partial^\nu \Phi^{\mathrm{A}} - g^{\mu\nu} \mathscr{L},$$

Stress-Energy Tensor

Energy of the ψ field in interaction

- *Stress-Energy tensor* is defined as

$$T^{\mu\nu} = \frac{\partial \mathscr{L}}{\partial(\partial_\mu \Phi^{\mathrm{A}})}\partial^\nu \Phi^{\mathrm{A}} - g^{\mu\nu}\mathscr{L},$$

$$\Rightarrow \quad T^{\mu\nu} = \frac{\partial \mathscr{L}_{\psi+int}}{\partial(\partial_\mu \psi)}\partial^\nu \psi + \frac{\partial \mathscr{L}_{\psi+int}}{\partial(\partial_\mu \overline{\psi})}\partial^\nu \overline{\psi} - g^{\mu\nu}\mathscr{L}_{\psi+int}$$

Stress-Energy Tensor

Energy of the ψ field in interaction

- *Stress-Energy tensor* is defined as

$$T^{\mu\nu} = \frac{\partial \mathscr{L}}{\partial(\partial_\mu \Phi^{\mathrm{A}})}\partial^\nu \Phi^{\mathrm{A}} - g^{\mu\nu}\mathscr{L},$$

$$\Rightarrow \quad T^{\mu\nu} = \frac{\partial \mathscr{L}_{\psi+int}}{\partial(\partial_\mu \psi)}\partial^\nu \psi + \frac{\partial \mathscr{L}_{\psi+int}}{\partial(\partial_\mu \overline{\psi})}\partial^\nu \overline{\psi} - g^{\mu\nu}\mathscr{L}_{\psi+int}$$

$$\Rightarrow \quad \mathscr{H}_{\psi+int} = T^{00} = \frac{\partial \mathscr{L}_{\psi+int}}{\partial(\partial_0 \psi)}\partial^0 \psi + \frac{\partial \mathscr{L}_{\psi+int}}{\partial(\partial_0 \overline{\psi})}\partial^0 \overline{\psi} - \mathscr{L}_{\psi+int}$$

Stress-Energy Tensor

Energy of the ψ field in interaction

- *Stress-Energy tensor* is defined as

$$T^{\mu\nu} = \frac{\partial \mathscr{L}}{\partial(\partial_\mu \Phi^{\mathrm{A}})}\partial^\nu \Phi^{\mathrm{A}} - g^{\mu\nu}\mathscr{L},$$

$$\Rightarrow \quad T^{\mu\nu} = \frac{\partial \mathscr{L}_{\psi+int}}{\partial(\partial_\mu \psi)}\partial^\nu \psi + \frac{\partial \mathscr{L}_{\psi+int}}{\partial(\partial_\mu \overline{\psi})}\partial^\nu \overline{\psi} - g^{\mu\nu}\mathscr{L}_{\psi+int}$$

$$\Rightarrow \quad \mathscr{H}_{\psi+int} = T^{00} = \frac{\partial \mathscr{L}_{\psi+int}}{\partial(\partial_0 \psi)}\partial^0 \psi + \frac{\partial \mathscr{L}_{\psi+int}}{\partial(\partial_0 \overline{\psi})}\partial^0 \overline{\psi} - \mathscr{L}_{\psi+int}$$

Now, $\mathscr{L}_{\psi+int} = \overline{\psi}\left[i\gamma^\mu\left(\partial_\mu + i\mathrm{g}_\omega \omega_\mu\right) - \left(m - \mathrm{g}_\sigma \sigma\right)\right]\psi$

Stress-Energy Tensor

Energy of the ψ field in interaction

- *Stress-Energy tensor* is defined as

$$T^{\mu\nu} = \frac{\partial \mathscr{L}}{\partial(\partial_\mu \Phi^{\mathrm{A}})}\partial^\nu \Phi^{\mathrm{A}} - g^{\mu\nu}\mathscr{L},$$

$$\Rightarrow \quad T^{\mu\nu} = \frac{\partial \mathscr{L}_{\psi+int}}{\partial(\partial_\mu \psi)}\partial^\nu \psi + \frac{\partial \mathscr{L}_{\psi+int}}{\partial(\partial_\mu \overline{\psi})}\partial^\nu \overline{\psi} - g^{\mu\nu}\mathscr{L}_{\psi+int}$$

$$\Rightarrow \quad \mathscr{H}_{\psi+int} = T^{00} = \frac{\partial \mathscr{L}_{\psi+int}}{\partial(\partial_0 \psi)}\partial^0 \psi + \frac{\partial \mathscr{L}_{\psi+int}}{\partial(\partial_0 \overline{\psi})}\partial^0 \overline{\psi} - \mathscr{L}_{\psi+int}$$

$$\text{Now, } \mathscr{L}_{\psi+int} = \overline{\psi}\left[i\gamma^\mu\left(\partial_\mu + i\mathrm{g}_\omega \omega_\mu\right) - \left(m - \mathrm{g}_\sigma \sigma\right)\right]\psi$$

$$\Rightarrow \quad \mathscr{H}_{\psi+int} = \langle \overline{\psi} i\gamma^0 \partial_0 \psi \rangle$$

Stress-Energy Tensor

Energy of the ψ field in interaction

- *Stress-Energy tensor* is defined as

$$T^{\mu\nu} = \frac{\partial \mathscr{L}}{\partial(\partial_\mu \Phi^{\mathrm{A}})}\partial^\nu \Phi^{\mathrm{A}} - g^{\mu\nu}\mathscr{L},$$

$$\Rightarrow \quad T^{\mu\nu} = \frac{\partial \mathscr{L}_{\psi+int}}{\partial(\partial_\mu \psi)}\partial^\nu \psi + \frac{\partial \mathscr{L}_{\psi+int}}{\partial(\partial_\mu \overline{\psi})}\partial^\nu \overline{\psi} - g^{\mu\nu}\mathscr{L}_{\psi+int}$$

$$\Rightarrow \quad \mathscr{H}_{\psi+int} = T^{00} = \frac{\partial \mathscr{L}_{\psi+int}}{\partial(\partial_0 \psi)}\partial^0 \psi + \frac{\partial \mathscr{L}_{\psi+int}}{\partial(\partial_0 \overline{\psi})}\partial^0 \overline{\psi} - \mathscr{L}_{\psi+int}$$

$$\text{Now, } \mathscr{L}_{\psi+int} = \overline{\psi}\left[i\gamma^\mu\left(\partial_\mu + i\mathrm{g}_\omega \omega_\mu\right) - \left(m - \mathrm{g}_\sigma \sigma\right)\right]\psi$$

$$\Rightarrow \quad \mathscr{H}_{\psi+int} = \langle \overline{\psi} i\gamma^0 \partial_0 \psi \rangle = \langle \overline{\psi}\gamma_0 p_0 \psi \rangle$$

Stress-Energy Tensor

Energy of the ψ field in interaction

- *Stress-Energy tensor* is defined as

$$T^{\mu\nu} = \frac{\partial \mathscr{L}}{\partial(\partial_\mu \Phi^{\mathrm{A}})}\partial^\nu \Phi^{\mathrm{A}} - g^{\mu\nu}\mathscr{L},$$

$$\Rightarrow \quad T^{\mu\nu} = \frac{\partial \mathscr{L}_{\psi+int}}{\partial(\partial_\mu \psi)}\partial^\nu \psi + \frac{\partial \mathscr{L}_{\psi+int}}{\partial(\partial_\mu \overline{\psi})}\partial^\nu \overline{\psi} - g^{\mu\nu}\mathscr{L}_{\psi+int}$$

$$\Rightarrow \quad \mathscr{H}_{\psi+int} = T^{00} = \frac{\partial \mathscr{L}_{\psi+int}}{\partial(\partial_0 \psi)}\partial^0 \psi + \frac{\partial \mathscr{L}_{\psi+int}}{\partial(\partial_0 \overline{\psi})}\partial^0 \overline{\psi} - \mathscr{L}_{\psi+int}$$

$$\text{Now, } \mathscr{L}_{\psi+int} = \overline{\psi}\left[i\gamma^\mu\left(\partial_\mu + i\mathrm{g}_\omega \omega_\mu\right) - \left(m - \mathrm{g}_\sigma \sigma\right)\right]\psi$$

$$\Rightarrow \quad \mathscr{H}_{\psi+int} = \langle \overline{\psi} i\gamma^0 \partial_0 \psi \rangle = \langle \overline{\psi}\gamma_0 p_0 \psi \rangle$$

- Remember, $e(\boldsymbol{p}) \equiv p_0(\boldsymbol{p}) = k_0 + \mathrm{g}_\omega \omega_0$

Stress-Energy Tensor

Energy of the ψ field in interaction

- *Stress-Energy tensor* is defined as

$$T^{\mu\nu} = \frac{\partial \mathscr{L}}{\partial(\partial_\mu \Phi^{\mathrm{A}})}\partial^\nu \Phi^{\mathrm{A}} - g^{\mu\nu}\mathscr{L},$$

$$\Rightarrow \quad T^{\mu\nu} = \frac{\partial \mathscr{L}_{\psi+int}}{\partial(\partial_\mu \psi)}\partial^\nu \psi + \frac{\partial \mathscr{L}_{\psi+int}}{\partial(\partial_\mu \overline{\psi})}\partial^\nu \overline{\psi} - g^{\mu\nu}\mathscr{L}_{\psi+int}$$

$$\Rightarrow \quad \mathscr{H}_{\psi+int} = T^{00} = \frac{\partial \mathscr{L}_{\psi+int}}{\partial(\partial_0 \psi)}\partial^0 \psi + \frac{\partial \mathscr{L}_{\psi+int}}{\partial(\partial_0 \overline{\psi})}\partial^0 \overline{\psi} - \mathscr{L}_{\psi+int}$$

$$\text{Now, } \mathscr{L}_{\psi+int} = \overline{\psi}\left[i\gamma^\mu\left(\partial_\mu + i\mathrm{g}_\omega \omega_\mu\right) - \left(m - \mathrm{g}_\sigma \sigma\right)\right]\psi$$

$$\Rightarrow \quad \mathscr{H}_{\psi+int} = \langle \overline{\psi} i\gamma^0 \partial_0 \psi \rangle = \langle \overline{\psi}\gamma_0 p_0 \psi \rangle$$

- Remember, $e(\boldsymbol{p}) \equiv p_0(\boldsymbol{p}) = k_0 + \mathrm{g}_\omega \omega_0 = \mathrm{g}_\omega \omega_0 + \sqrt{\boldsymbol{p}^2 + (m - \mathrm{g}_\sigma \sigma)^2}$,

Stress-Energy Tensor

Energy of the ψ field in interaction

- *Stress-Energy tensor* is defined as

$$T^{\mu\nu} = \frac{\partial \mathscr{L}}{\partial(\partial_\mu \Phi^{\mathrm{A}})}\partial^\nu \Phi^{\mathrm{A}} - g^{\mu\nu}\mathscr{L},$$

$$\Rightarrow \quad T^{\mu\nu} = \frac{\partial \mathscr{L}_{\psi+int}}{\partial(\partial_\mu \psi)}\partial^\nu \psi + \frac{\partial \mathscr{L}_{\psi+int}}{\partial(\partial_\mu \overline{\psi})}\partial^\nu \overline{\psi} - g^{\mu\nu}\mathscr{L}_{\psi+int}$$

$$\Rightarrow \quad \mathscr{H}_{\psi+int} = T^{00} = \frac{\partial \mathscr{L}_{\psi+int}}{\partial(\partial_0 \psi)}\partial^0 \psi + \frac{\partial \mathscr{L}_{\psi+int}}{\partial(\partial_0 \overline{\psi})}\partial^0 \overline{\psi} - \mathscr{L}_{\psi+int}$$

$$\text{Now, } \mathscr{L}_{\psi+int} = \overline{\psi}\left[i\gamma^\mu\left(\partial_\mu + i\mathrm{g}_\omega \omega_\mu\right) - \left(m - \mathrm{g}_\sigma \sigma\right)\right]\psi$$

$$\Rightarrow \quad \mathscr{H}_{\psi+int} = \langle \overline{\psi} i\gamma^0 \partial_0 \psi\rangle = \langle \overline{\psi}\gamma_0 p_0 \psi\rangle$$

- Remember, $e(\boldsymbol{p}) \equiv p_0(\boldsymbol{p}) = k_0 + \mathrm{g}_\omega \omega_0 = \mathrm{g}_\omega \omega_0 + \sqrt{\boldsymbol{p}^2 + (m - \mathrm{g}_\sigma \sigma)^2}$, and

$$\langle \overline{\psi}\Gamma\psi\rangle = \sum_\kappa \int \frac{d\boldsymbol{p}}{(2\pi)^3}(\overline{\psi}\Gamma\psi)_{\boldsymbol{p}_\kappa}\Theta[\mu - \mathrm{e}(\boldsymbol{p})].$$

Stress-Energy Tensor

Total energy density

Stress-Energy Tensor

Total energy density

- So we can write

$$\langle \overline{\psi} \gamma_0 p_0 \psi \rangle = \frac{2}{\pi^2} \int_0^{k_f} p^2 \, dp \; e(p)$$

Stress-Energy Tensor

Total energy density

- So we can write

$$\begin{aligned}\langle\overline{\psi}\gamma_0 p_0\psi\rangle &= \frac{2}{\pi^2}\int_0^{k_f} p^2\,dp\;e(p)\\ &= \frac{2}{\pi^2}\int_0^{k_f} p^2\,dp\left[\mathrm{g}_\omega\omega_0+\sqrt{p^2+(m-\mathrm{g}_\sigma\sigma)^2}\right]\end{aligned}$$

Stress-Energy Tensor

Total energy density

- So we can write

$$\begin{aligned}
\langle \overline{\psi}\gamma_0 p_0 \psi \rangle &= \frac{2}{\pi^2}\int_0^{k_f} p^2\, dp\ e(p) \\
&= \frac{2}{\pi^2}\int_0^{k_f} p^2\, dp \left[\mathrm{g}_\omega \omega_0 + \sqrt{p^2 + (m - \mathrm{g}_\sigma \sigma)^2} \right] \\
&= \mathrm{g}_\omega \omega_0 \rho_b + \frac{2}{\pi^2}\int_0^{k_f} p^2\, dp \sqrt{p^2 + (m - \mathrm{g}_\sigma \sigma)^2}
\end{aligned}$$

Stress-Energy Tensor

Total energy density

- So we can write

$$
\begin{aligned}
\langle \overline{\psi}\gamma_0 p_0 \psi \rangle &= \frac{2}{\pi^2}\int_0^{k_f} p^2\, dp\ e(p) \\
&= \frac{2}{\pi^2}\int_0^{k_f} p^2\, dp \left[\mathrm{g}_\omega \omega_0 + \sqrt{p^2 + (m - \mathrm{g}_\sigma \sigma)^2} \right] \\
&= \mathrm{g}_\omega \omega_0 \rho_b + \frac{2}{\pi^2}\int_0^{k_f} p^2\, dp \sqrt{p^2 + (m - \mathrm{g}_\sigma \sigma)^2} \\
&= m_\omega^2 \omega_0^2 + \frac{2}{\pi^2}\int_0^{k_f} p^2\, dp \sqrt{p^2 + (m - \mathrm{g}_\sigma \sigma)^2}
\end{aligned}
$$

Stress-Energy Tensor

Total energy density

- So we can write

$$\begin{aligned}
\langle \overline{\psi}\gamma_0 p_0 \psi \rangle &= \frac{2}{\pi^2}\int_0^{k_f} p^2\, dp\ e(p) \\
&= \frac{2}{\pi^2}\int_0^{k_f} p^2\, dp \left[\mathrm{g}_\omega \omega_0 + \sqrt{p^2 + (m - \mathrm{g}_\sigma \sigma)^2} \right] \\
&= \mathrm{g}_\omega \omega_0 \rho_b + \frac{2}{\pi^2}\int_0^{k_f} p^2\, dp \sqrt{p^2 + (m - \mathrm{g}_\sigma \sigma)^2} \\
&= m_\omega^2 \omega_0^2 + \frac{2}{\pi^2}\int_0^{k_f} p^2\, dp \sqrt{p^2 + (m - \mathrm{g}_\sigma \sigma)^2}
\end{aligned}$$

- So the energy density of the $\sigma - \omega$ model is given by

$$\epsilon = \mathscr{H}_\sigma + \mathscr{H}_\omega + \mathscr{H}_{\psi + int}$$

Stress-Energy Tensor

Total energy density

- So we can write

$$\begin{aligned}
\langle \overline{\psi}\gamma_0 p_0 \psi \rangle &= \frac{2}{\pi^2}\int_0^{k_f} p^2\, dp\ e(p) \\
&= \frac{2}{\pi^2}\int_0^{k_f} p^2\, dp \left[\mathrm{g}_\omega \omega_0 + \sqrt{p^2 + (m - \mathrm{g}_\sigma \sigma)^2} \right] \\
&= \mathrm{g}_\omega \omega_0 \rho_b + \frac{2}{\pi^2}\int_0^{k_f} p^2\, dp \sqrt{p^2 + (m - \mathrm{g}_\sigma \sigma)^2} \\
&= m_\omega^2 \omega_0^2 + \frac{2}{\pi^2}\int_0^{k_f} p^2\, dp \sqrt{p^2 + (m - \mathrm{g}_\sigma \sigma)^2}
\end{aligned}$$

- So the energy density of the $\sigma - \omega$ model is given by

$$\epsilon = \mathscr{H}_\sigma + \mathscr{H}_\omega + \mathscr{H}_{\psi + int} = \frac{1}{2} m_\sigma^2 \sigma^2 - \frac{1}{2} m_\omega^2 \omega_0^2 + \langle \overline{\psi}\gamma_0 p_0 \psi \rangle$$

Stress-Energy Tensor

Total energy density

- So we can write

$$\begin{aligned}
\langle \overline{\psi}\gamma_0 p_0 \psi \rangle &= \frac{2}{\pi^2}\int_0^{k_f} p^2\, dp\ e(p) \\
&= \frac{2}{\pi^2}\int_0^{k_f} p^2\, dp \left[g_\omega \omega_0 + \sqrt{p^2 + (m - g_\sigma \sigma)^2} \right] \\
&= g_\omega \omega_0 \rho_b + \frac{2}{\pi^2}\int_0^{k_f} p^2\, dp \sqrt{p^2 + (m - g_\sigma \sigma)^2} \\
&= m_\omega^2 \omega_0^2 + \frac{2}{\pi^2}\int_0^{k_f} p^2\, dp \sqrt{p^2 + (m - g_\sigma \sigma)^2}
\end{aligned}$$

- So the energy density of the $\sigma - \omega$ model is given by

$$\begin{aligned}
\epsilon &= \mathscr{H}_\sigma + \mathscr{H}_\omega + \mathscr{H}_{\psi + int} = \frac{1}{2} m_\sigma^2 \sigma^2 - \frac{1}{2} m_\omega^2 \omega_0^2 + \langle \overline{\psi}\gamma_0 p_0 \psi \rangle \\
&= \boxed{\frac{1}{2} m_\sigma^2 \sigma^2 + \frac{1}{2} m_\omega^2 \omega_0^2 + \frac{2}{\pi^2}\int_0^{k_f} p^2\, dp \sqrt{p^2 + (m - g_\sigma \sigma)^2}}.
\end{aligned}$$

Stress-Energy Tensor

Pressure

Stress-Energy Tensor

Pressure

- **Problem 1: Starting from the definition of Stress-Energy tensor, show that the pressure in $\sigma - \omega$ model is given by,**

$$\mathsf{p} = \frac{1}{3}\langle\overline{\psi}\gamma_i p_i \psi\rangle + \mathscr{L}_\sigma + \mathscr{L}_\omega.$$

Stress-Energy Tensor

Pressure

- **Problem 1: Starting from the definition of Stress-Energy tensor, show that the pressure in $\sigma - \omega$ model is given by,**
$$\mathsf{p} = \frac{1}{3}\langle\overline{\psi}\gamma_i p_i \psi\rangle + \mathscr{L}_\sigma + \mathscr{L}_\omega.$$
- We can easily show from Lec-6 (see slide 13)

$$(\overline{\psi}\boldsymbol{\gamma}\psi)_{\boldsymbol{p}_\kappa} = \frac{\partial E(\boldsymbol{p})}{\partial \boldsymbol{p}},$$

Stress-Energy Tensor

Pressure

- **Problem 1: Starting from the definition of Stress-Energy tensor, show that the pressure in $\sigma - \omega$ model is given by,**
$$\mathsf{p} = \frac{1}{3}\langle\overline{\psi}\gamma_i p_i \psi\rangle + \mathscr{L}_\sigma + \mathscr{L}_\omega.$$
- We can easily show from Lec-6 (see slide 13)

$$(\overline{\psi}\boldsymbol{\gamma}\psi)_{\boldsymbol{p}_\kappa} = \frac{\partial E(\boldsymbol{p})}{\partial \boldsymbol{p}}, \quad \text{Recall, } E(\boldsymbol{p}) = \sqrt{\boldsymbol{p}^2 + (m - \mathrm{g}_\sigma \sigma)^2}.$$

Stress-Energy Tensor

Pressure

- **Problem 1: Starting from the definition of Stress-Energy tensor, show that the pressure in $\sigma - \omega$ model is given by,**
$$\mathrm{p} = \frac{1}{3}\langle\overline{\psi}\gamma_i p_i \psi\rangle + \mathscr{L}_\sigma + \mathscr{L}_\omega.$$
- We can easily show from Lec-6 (see slide 13)

$$(\overline{\psi}\boldsymbol{\gamma}\psi)_{\boldsymbol{p}_\kappa} = \frac{\partial E(\boldsymbol{p})}{\partial \boldsymbol{p}}, \quad \text{Recall, } E(\boldsymbol{p}) = \sqrt{\boldsymbol{p}^2 + (m - \mathrm{g}_\sigma \sigma)^2}.$$

$$\Rightarrow \quad \langle\overline{\psi}\boldsymbol{\gamma}\cdot\boldsymbol{p}\psi\rangle = \frac{2}{\pi^2}\int_0^{k_f} p^2\, dp \frac{p^2}{\sqrt{p^2 + (m - \mathrm{g}_\sigma \sigma)^2}}$$

Stress-Energy Tensor

Pressure

- **Problem 1: Starting from the definition of Stress-Energy tensor, show that the pressure in $\sigma - \omega$ model is given by,**
$$\mathtt{p} = \tfrac{1}{3}\langle\overline{\psi}\gamma_i p_i \psi\rangle + \mathscr{L}_\sigma + \mathscr{L}_\omega.$$
- We can easily show from Lec-6 (see slide 13)

$$(\overline{\psi}\boldsymbol{\gamma}\psi)_{\boldsymbol{p}_\kappa} = \frac{\partial E(\boldsymbol{p})}{\partial \boldsymbol{p}}, \quad \text{Recall, } E(\boldsymbol{p}) = \sqrt{\boldsymbol{p}^2 + (m - \mathrm{g}_\sigma\sigma)^2}.$$

$$\Rightarrow \quad \langle\overline{\psi}\boldsymbol{\gamma}\cdot\boldsymbol{p}\psi\rangle = \frac{2}{\pi^2}\int_0^{k_f} p^2\,dp \frac{p^2}{\sqrt{p^2 + (m - \mathrm{g}_\sigma\sigma)^2}}$$

$$\Rightarrow \quad \boxed{\mathtt{p} = -\frac{1}{2}m_\sigma^2\sigma^2 + \frac{1}{2}m_\omega^2\omega_0^2 + \frac{1}{3}\frac{2}{\pi^2}\int_0^{k_f} p^2\,dp \frac{p^2}{\sqrt{p^2 + (m - \mathrm{g}_\sigma\sigma)^2}}}.$$

under all the approximations discussed.

Stress-Energy Tensor

Pressure

- **Problem 1: Starting from the definition of Stress-Energy tensor, show that the pressure in $\sigma - \omega$ model is given by,**
$$\mathtt{p} = \tfrac{1}{3}\langle\overline{\psi}\gamma_i p_i \psi\rangle + \mathscr{L}_\sigma + \mathscr{L}_\omega.$$
- We can easily show from Lec-6 (see slide 13)

$$(\overline{\psi}\boldsymbol{\gamma}\psi)_{\boldsymbol{p}_\kappa} = \frac{\partial E(\boldsymbol{p})}{\partial \boldsymbol{p}}, \quad \text{Recall, } E(\boldsymbol{p}) = \sqrt{\boldsymbol{p}^2 + (m - \mathrm{g}_\sigma\sigma)^2}.$$

$$\Rightarrow \quad \langle\overline{\psi}\boldsymbol{\gamma}\cdot\boldsymbol{p}\psi\rangle = \frac{2}{\pi^2}\int_0^{k_f} p^2\,dp \frac{p^2}{\sqrt{p^2 + (m - \mathrm{g}_\sigma\sigma)^2}}$$

$$\Rightarrow \quad \boxed{\mathtt{p} = -\frac{1}{2}m_\sigma^2\sigma^2 + \frac{1}{2}m_\omega^2\omega_0^2 + \frac{1}{3}\frac{2}{\pi^2}\int_0^{k_f} p^2\,dp \frac{p^2}{\sqrt{p^2 + (m - \mathrm{g}_\sigma\sigma)^2}}}.$$

under all the approximations discussed.

- $m_\sigma, m_\omega, \mathrm{g}_\sigma, \mathrm{g}_\omega, m$ are provided. For a given ρ_b solve the field equations self consistently and then find ϵ and $\mathtt{p}$.

Problem with K_0 and J

Self interactions of σ field and ρ_μ meson

Problem with K_0 and J

Self interactions of σ field and ρ_μ meson

- Problem to reproduce data on incompressibility was found.

Problem with K_0 and J

Self interactions of σ field and ρ_μ meson

- Problem to reproduce data on incompressibility was found.
- The remedy was suggested in the form of the Lagrangian

$$U(\sigma) = \frac{1}{3}g_2\sigma^3 + \frac{1}{4}g_3\sigma^4$$

Problem with K_0 and J

Self interactions of σ field and ρ_μ meson

- Problem to reproduce data on incompressibility was found.
- The remedy was suggested in the form of the Lagrangian

$$U(\sigma) = \frac{1}{3}g_2\sigma^3 + \frac{1}{4}g_3\sigma^4$$

$$\Rightarrow \quad \mathscr{L} = \mathscr{L}_\sigma + \mathscr{L}_\omega + \mathscr{L}_\psi + \mathscr{L}_{int} - U(\sigma)$$

Problem with K_0 and J

Self interactions of σ field and ρ_μ meson

- Problem to reproduce data on incompressibility was found.
- The remedy was suggested in the form of the Lagrangian

$$U(\sigma) = \frac{1}{3}g_2\sigma^3 + \frac{1}{4}g_3\sigma^4$$
$$\Rightarrow \quad \mathscr{L} = \mathscr{L}_\sigma + \mathscr{L}_\omega + \mathscr{L}_\psi + \mathscr{L}_{int} - U(\sigma)$$

- Similar problem appeared for J when calculations were extended for asymmetric matter.

Problem with K_0 and J

Self interactions of σ field and ρ_μ meson

- Problem to reproduce data on incompressibility was found.
- The remedy was suggested in the form of the Lagrangian

$$U(\sigma) = \frac{1}{3}g_2\sigma^3 + \frac{1}{4}g_3\sigma^4$$
$$\Rightarrow \quad \mathscr{L} = \mathscr{L}_\sigma + \mathscr{L}_\omega + \mathscr{L}_\psi + \mathscr{L}_{int} - U(\sigma)$$

- Similar problem appeared for J when calculations were extended for asymmetric matter. Necessity of a isospin sensitive meson was realized.

Problem with K_0 and J

Self interactions of σ field and ρ_μ meson

- Problem to reproduce data on incompressibility was found.
- The remedy was suggested in the form of the Lagrangian

$$U(\sigma) = \frac{1}{3}g_2\sigma^3 + \frac{1}{4}g_3\sigma^4$$
$$\Rightarrow \quad \mathscr{L} = \mathscr{L}_\sigma + \mathscr{L}_\omega + \mathscr{L}_\psi + \mathscr{L}_{int} - U(\sigma)$$

- Similar problem appeared for J when calculations were extended for asymmetric matter. Necessity of a isospin sensitive meson was realized. But it needs to reproduce a *Lorentz* scalar in the Lagrangian.

Problem with K_0 and J

Self interactions of σ field and ρ_μ meson

- Problem to reproduce data on incompressibility was found.
- The remedy was suggested in the form of the Lagrangian

$$U(\sigma) = \frac{1}{3}g_2\sigma^3 + \frac{1}{4}g_3\sigma^4$$
$$\Rightarrow \quad \mathscr{L} = \mathscr{L}_\sigma + \mathscr{L}_\omega + \mathscr{L}_\psi + \mathscr{L}_{int} - U(\sigma)$$

- Similar problem appeared for J when calculations were extended for asymmetric matter. Necessity of a isospin sensitive meson was realized. But it needs to reproduce a *Lorentz* scalar in the Lagrangian.
- **Problem 2: Show that Dirac Lagrangian is invariant under the isospin transformation $\psi \longrightarrow e^{-i\boldsymbol{\tau}\cdot\Lambda/2}\psi$, where, $\boldsymbol{\tau}$ are the Pauli matrices and $\boldsymbol{\Lambda}$ is an arbitrary constant vector in isospin space. Show further that corresponding conserved isospin Noether current is $\frac{1}{2}\overline{\psi}\gamma^\mu\boldsymbol{\tau}\psi$.**

Problem with K_0 and J

Self interactions of σ field and $\boldsymbol{\rho}_\mu$ meson

- Problem to reproduce data on incompressibility was found.
- The remedy was suggested in the form of the Lagrangian

$$U(\sigma) = \frac{1}{3}\mathrm{g}_2\sigma^3 + \frac{1}{4}\mathrm{g}_3\sigma^4$$
$$\Rightarrow \quad \mathscr{L} = \mathscr{L}_\sigma + \mathscr{L}_\omega + \mathscr{L}_\psi + \mathscr{L}_{int} - U(\sigma)$$

- Similar problem appeared for J when calculations were extended for asymmetric matter. Necessity of a isospin sensitive meson was realized. But it needs to reproduce a *Lorentz* scalar in the Lagrangian.
- **Problem 2: Show that Dirac Lagrangian is invariant under the isospin transformation $\psi \longrightarrow e^{-i\boldsymbol{\tau}\cdot\Lambda/2}\psi$, where, $\boldsymbol{\tau}$ are the Pauli matrices and $\boldsymbol{\Lambda}$ is an arbitrary constant vector in isospin space. Show further that corresponding conserved isospin Noether current is $\frac{1}{2}\overline{\psi}\gamma^\mu\boldsymbol{\tau}\psi$.**
- We couple this to a *isovector-vector* meson $\boldsymbol{\rho}_\mu$ with coupling constant $\mathrm{g}_{\boldsymbol{\rho}}$ as

$$\mathscr{L}_{int} = \mathrm{g}_\sigma\overline{\psi}\psi\sigma - \mathrm{g}_\omega\overline{\psi}\gamma^\mu\psi\omega_\mu - \frac{1}{2}\mathrm{g}_{\boldsymbol{\rho}}\overline{\psi}\gamma^\mu\boldsymbol{\tau}\psi\cdot\boldsymbol{\rho}_\mu.$$

Lagrangian of free ρ_μ meson

Field equation of ρ_μ meson

Lagrangian of free $\boldsymbol{\rho}_\mu$ meson

Field equation of $\boldsymbol{\rho}_\mu$ meson

- The Lagrangian is given by

$$\mathscr{L}_{\boldsymbol{\rho}} = -\frac{1}{4}\mathbf{R}_{\mu\nu}\mathbf{R}^{\mu\nu} + \frac{1}{2}m_{\boldsymbol{\rho}}^2\boldsymbol{\rho}_\mu\boldsymbol{\rho}^\mu,$$

Lagrangian of free $\boldsymbol{\rho}_\mu$ meson

Field equation of $\boldsymbol{\rho}_\mu$ meson

- The Lagrangian is given by

$$\begin{aligned} \mathscr{L}_{\boldsymbol{\rho}} &= -\frac{1}{4}\mathbf{R}_{\mu\nu}\mathbf{R}^{\mu\nu} + \frac{1}{2}m_{\boldsymbol{\rho}}^2\boldsymbol{\rho}_\mu\boldsymbol{\rho}^\mu, \\ \text{where, } \mathbf{R}_{\mu\nu} &= \partial_\mu\boldsymbol{\rho}_\nu - \partial_\nu\boldsymbol{\rho}_\mu + \mathrm{g}_{\boldsymbol{\rho}}(\boldsymbol{\rho}_\mu \times \boldsymbol{\rho}_\nu) \end{aligned}$$

Lagrangian of free $\boldsymbol{\rho}_\mu$ meson

Field equation of $\boldsymbol{\rho}_\mu$ meson

- The Lagrangian is given by

$$\begin{aligned} \mathscr{L}_{\boldsymbol{\rho}} &= -\frac{1}{4}\mathbf{R}_{\mu\nu}\mathbf{R}^{\mu\nu} + \frac{1}{2}m_{\boldsymbol{\rho}}^2\boldsymbol{\rho}_\mu\boldsymbol{\rho}^\mu, \\ \text{where, } \mathbf{R}_{\mu\nu} &= \partial_\mu\boldsymbol{\rho}_\nu - \partial_\nu\boldsymbol{\rho}_\mu + \mathrm{g}_{\boldsymbol{\rho}}(\boldsymbol{\rho}_\mu \times \boldsymbol{\rho}_\nu) \end{aligned}$$

- Going through the same procedure as ω_μ meson we find the field equation for $\boldsymbol{\rho}_\mu$ meson as

$$\partial_\mu\mathbf{R}^{\mu\nu} + m_{\boldsymbol{\rho}}^2\boldsymbol{\rho}^\nu = \frac{1}{2}\mathrm{g}_{\boldsymbol{\rho}}\overline{\psi}\gamma^\nu\boldsymbol{\tau}\psi$$

Lagrangian of free $\boldsymbol{\rho}_\mu$ meson

Field equation of $\boldsymbol{\rho}_\mu$ meson

- The Lagrangian is given by

$$\begin{aligned} \mathscr{L}_{\boldsymbol{\rho}} &= -\frac{1}{4}\mathbf{R}_{\mu\nu}\mathbf{R}^{\mu\nu} + \frac{1}{2}m_{\boldsymbol{\rho}}^2\boldsymbol{\rho}_\mu\boldsymbol{\rho}^\mu, \\ \text{where, } \mathbf{R}_{\mu\nu} &= \partial_\mu\boldsymbol{\rho}_\nu - \partial_\nu\boldsymbol{\rho}_\mu + \mathrm{g}_{\boldsymbol{\rho}}(\boldsymbol{\rho}_\mu \times \boldsymbol{\rho}_\nu) \end{aligned}$$

- Going through the same procedure as ω_μ meson we find the field equation for $\boldsymbol{\rho}_\mu$ meson as

$$\partial_\mu\mathbf{R}^{\mu\nu} + m_{\boldsymbol{\rho}}^2\boldsymbol{\rho}^\nu = \frac{1}{2}\mathrm{g}_{\boldsymbol{\rho}}\overline{\psi}\gamma^\nu\boldsymbol{\tau}\psi$$

- Only 3-component of the $\boldsymbol{\rho}_\mu$ meson has finite mean value!

Lagrangian of free $\boldsymbol{\rho}_\mu$ meson

Field equation of $\boldsymbol{\rho}_\mu$ meson

- The Lagrangian is given by

$$\begin{aligned} \mathscr{L}_{\boldsymbol{\rho}} &= -\frac{1}{4}\mathbf{R}_{\mu\nu}\mathbf{R}^{\mu\nu} + \frac{1}{2}m_{\boldsymbol{\rho}}^2\boldsymbol{\rho}_\mu\boldsymbol{\rho}^\mu, \\ \text{where, } \mathbf{R}_{\mu\nu} &= \partial_\mu\boldsymbol{\rho}_\nu - \partial_\nu\boldsymbol{\rho}_\mu + \mathrm{g}_{\boldsymbol{\rho}}(\boldsymbol{\rho}_\mu \times \boldsymbol{\rho}_\nu) \end{aligned}$$

- Going through the same procedure as ω_μ meson we find the field equation for $\boldsymbol{\rho}_\mu$ meson as

$$\partial_\mu\mathbf{R}^{\mu\nu} + m_{\boldsymbol{\rho}}^2\boldsymbol{\rho}^\nu = \frac{1}{2}\mathrm{g}_{\boldsymbol{\rho}}\overline{\psi}\gamma^\nu\boldsymbol{\tau}\psi$$

- Only 3-component of the $\boldsymbol{\rho}_\mu$ meson has finite mean value!

$$\partial_\mu\mathrm{R}_3^{\mu\nu} + m_{\boldsymbol{\rho}}^2\rho_3^\nu = \frac{1}{2}\mathrm{g}_{\boldsymbol{\rho}}\overline{\psi}\gamma^\nu\tau_3\psi$$

Lagrangian of free $\boldsymbol{\rho}_\mu$ meson

Field equation of $\boldsymbol{\rho}_\mu$ meson

- The Lagrangian is given by

$$\begin{aligned} \mathscr{L}_{\boldsymbol{\rho}} &= -\frac{1}{4}\mathbf{R}_{\mu\nu}\mathbf{R}^{\mu\nu} + \frac{1}{2}m_{\boldsymbol{\rho}}^2\boldsymbol{\rho}_\mu\boldsymbol{\rho}^\mu, \\ \text{where, } \mathbf{R}_{\mu\nu} &= \partial_\mu\boldsymbol{\rho}_\nu - \partial_\nu\boldsymbol{\rho}_\mu + \mathrm{g}_{\boldsymbol{\rho}}(\boldsymbol{\rho}_\mu \times \boldsymbol{\rho}_\nu) \end{aligned}$$

- Going through the same procedure as ω_μ meson we find the field equation for $\boldsymbol{\rho}_\mu$ meson as

$$\partial_\mu\mathbf{R}^{\mu\nu} + m_{\boldsymbol{\rho}}^2\boldsymbol{\rho}^\nu = \frac{1}{2}\mathrm{g}_{\boldsymbol{\rho}}\overline{\psi}\gamma^\nu\boldsymbol{\tau}\psi$$

- Only 3-component of the $\boldsymbol{\rho}_\mu$ meson has finite mean value!

$$\partial_\mu\mathrm{R}_3^{\mu\nu} + m_{\boldsymbol{\rho}}^2\rho_3^\nu = \frac{1}{2}\mathrm{g}_{\boldsymbol{\rho}}\overline{\psi}\gamma^\nu\tau_3\psi$$

$$\Rightarrow \quad (\Box + m_{\boldsymbol{\rho}}^2)\rho_3^\nu = \frac{1}{2}\mathrm{g}_{\boldsymbol{\rho}}\overline{\psi}\gamma^\nu\tau_3\psi, \text{ under charge symmetry.}$$

Lagrangian of free $\boldsymbol{\rho}_\mu$ meson

Field equation of $\boldsymbol{\rho}_\mu$ meson

- The Lagrangian is given by

$$\begin{aligned} \mathscr{L}_{\boldsymbol{\rho}} &= -\frac{1}{4}\mathbf{R}_{\mu\nu}\mathbf{R}^{\mu\nu} + \frac{1}{2}m_{\boldsymbol{\rho}}^2\boldsymbol{\rho}_\mu\boldsymbol{\rho}^\mu, \\ \text{where, } \mathbf{R}_{\mu\nu} &= \partial_\mu\boldsymbol{\rho}_\nu - \partial_\nu\boldsymbol{\rho}_\mu + \mathrm{g}_{\boldsymbol{\rho}}(\boldsymbol{\rho}_\mu \times \boldsymbol{\rho}_\nu) \end{aligned}$$

- Going through the same procedure as ω_μ meson we find the field equation for $\boldsymbol{\rho}_\mu$ meson as

$$\partial_\mu\mathbf{R}^{\mu\nu} + m_{\boldsymbol{\rho}}^2\boldsymbol{\rho}^\nu = \frac{1}{2}\mathrm{g}_{\boldsymbol{\rho}}\overline{\psi}\gamma^\nu\boldsymbol{\tau}\psi$$

- Only 3-component of the $\boldsymbol{\rho}_\mu$ meson has finite mean value!

$$\begin{aligned} &\partial_\mu\mathrm{R}_3^{\mu\nu} + m_{\boldsymbol{\rho}}^2\rho_3^\nu = \frac{1}{2}\mathrm{g}_{\boldsymbol{\rho}}\overline{\psi}\gamma^\nu\tau_3\psi \\ \Rightarrow\quad &(\Box + m_{\boldsymbol{\rho}}^2)\rho_3^\nu = \frac{1}{2}\mathrm{g}_{\boldsymbol{\rho}}\overline{\psi}\gamma^\nu\tau_3\psi, \text{ under charge symmetry.} \\ \Rightarrow\quad &(\Box + m_{\boldsymbol{\rho}}^2)\rho_3^0 = \frac{1}{2}\mathrm{g}_{\boldsymbol{\rho}}\overline{\psi}\gamma^0\tau_3\psi, \text{ under time reversal.} \end{aligned}$$

Lagrangian of free $\boldsymbol{\rho}_\mu$ meson

Field equation of $\boldsymbol{\rho}_\mu$ meson

- The Lagrangian is given by

$$\begin{aligned} \mathscr{L}_{\boldsymbol{\rho}} &= -\frac{1}{4}\mathbf{R}_{\mu\nu}\mathbf{R}^{\mu\nu} + \frac{1}{2}m_{\boldsymbol{\rho}}^2\boldsymbol{\rho}_\mu\boldsymbol{\rho}^\mu, \\ \text{where, } \mathbf{R}_{\mu\nu} &= \partial_\mu\boldsymbol{\rho}_\nu - \partial_\nu\boldsymbol{\rho}_\mu + \mathrm{g}_{\boldsymbol{\rho}}(\boldsymbol{\rho}_\mu \times \boldsymbol{\rho}_\nu) \end{aligned}$$

- Going through the same procedure as ω_μ meson we find the field equation for $\boldsymbol{\rho}_\mu$ meson as

$$\partial_\mu\mathbf{R}^{\mu\nu} + m_{\boldsymbol{\rho}}^2\boldsymbol{\rho}^\nu = \frac{1}{2}\mathrm{g}_{\boldsymbol{\rho}}\overline{\psi}\gamma^\nu\boldsymbol{\tau}\psi$$

- Only 3-component of the $\boldsymbol{\rho}_\mu$ meson has finite mean value!

$$\begin{aligned} & \partial_\mu\mathrm{R}_3^{\mu\nu} + m_{\boldsymbol{\rho}}^2\rho_3^\nu = \frac{1}{2}\mathrm{g}_{\boldsymbol{\rho}}\overline{\psi}\gamma^\nu\tau_3\psi \\ \Rightarrow \quad & (\Box + m_{\boldsymbol{\rho}}^2)\rho_3^\nu = \frac{1}{2}\mathrm{g}_{\boldsymbol{\rho}}\overline{\psi}\gamma^\nu\tau_3\psi, \text{ under charge symmetry.} \\ \Rightarrow \quad & (\Box + m_{\boldsymbol{\rho}}^2)\rho_3^0 = \frac{1}{2}\mathrm{g}_{\boldsymbol{\rho}}\overline{\psi}\gamma^0\tau_3\psi, \text{ under time reversal.} \\ \Rightarrow \quad & m_{\boldsymbol{\rho}}^2\langle\rho_3^0\rangle = \frac{1}{2}\mathrm{g}_{\boldsymbol{\rho}}\langle\overline{\psi}\gamma^0\tau_3\psi\rangle, \text{ static mean field.} \end{aligned}$$

Lagrangian of free ρ_μ meson

Field equation of $\boldsymbol{\rho}_\mu$ meson

- The Lagrangian is given by

$$\begin{aligned} \mathscr{L}_{\boldsymbol{\rho}} &= -\frac{1}{4}\mathbf{R}_{\mu\nu}\mathbf{R}^{\mu\nu} + \frac{1}{2}m_{\boldsymbol{\rho}}^2\boldsymbol{\rho}_\mu\boldsymbol{\rho}^\mu, \\ \text{where, } \mathbf{R}_{\mu\nu} &= \partial_\mu\boldsymbol{\rho}_\nu - \partial_\nu\boldsymbol{\rho}_\mu + \mathrm{g}_{\boldsymbol{\rho}}(\boldsymbol{\rho}_\mu \times \boldsymbol{\rho}_\nu) \end{aligned}$$

- Going through the same procedure as ω_μ meson we find the field equation for $\boldsymbol{\rho}_\mu$ meson as

$$\partial_\mu\mathbf{R}^{\mu\nu} + m_{\boldsymbol{\rho}}^2\boldsymbol{\rho}^\nu = \frac{1}{2}\mathrm{g}_{\boldsymbol{\rho}}\overline{\psi}\gamma^\nu\boldsymbol{\tau}\psi$$

- Only 3-component of the $\boldsymbol{\rho}_\mu$ meson has finite mean value!

$$\begin{aligned} & \partial_\mu \mathrm{R}_3^{\mu\nu} + m_{\boldsymbol{\rho}}^2\rho_3^\nu = \frac{1}{2}\mathrm{g}_{\boldsymbol{\rho}}\overline{\psi}\gamma^\nu\tau_3\psi \\ \Rightarrow \quad & (\Box + m_{\boldsymbol{\rho}}^2)\rho_3^\nu = \frac{1}{2}\mathrm{g}_{\boldsymbol{\rho}}\overline{\psi}\gamma^\nu\tau_3\psi, \text{ under charge symmetry.} \\ \Rightarrow \quad & (\Box + m_{\boldsymbol{\rho}}^2)\rho_3^0 = \frac{1}{2}\mathrm{g}_{\boldsymbol{\rho}}\overline{\psi}\gamma^0\tau_3\psi, \text{ under time reversal.} \\ \Rightarrow \quad & m_{\boldsymbol{\rho}}^2\langle\rho_3^0\rangle = \frac{1}{2}\mathrm{g}_{\boldsymbol{\rho}}\langle\overline{\psi}\gamma^0\tau_3\psi\rangle, \text{ static mean field.} \\ \Rightarrow \quad & \mathrm{g}_{\boldsymbol{\rho}}\langle\rho_3^0\rangle = \left(\frac{\mathrm{g}_{\boldsymbol{\rho}}}{m_{\boldsymbol{\rho}}}\right)^2\frac{(\rho_n - \rho_p)}{2}. \end{aligned}$$

Final field equations

Effect of $U(\sigma)$ and $\boldsymbol{\rho}_\mu$ meson

Final field equations

Effect of $U(\sigma)$ and $\boldsymbol{\rho}_\mu$ meson

- The field equations for σ, ω_μ and $\boldsymbol{\rho}_\mu$ fields are given by

$$m_\sigma^2 \langle \sigma \rangle = \mathrm{g}_\sigma \langle \overline{\psi}\psi \rangle = \mathrm{g}_\sigma \frac{2}{\pi^2} \int_0^{k_f} p^2 dp \frac{m - \mathrm{g}_\sigma \sigma}{\sqrt{p^2 + (m - \mathrm{g}_\sigma \sigma)^2}} - \mathrm{g}_2 \sigma^2 - \mathrm{g}_3 \sigma^3,$$

$$m_\omega^2 \langle \omega \rangle = \mathrm{g}_\omega \langle \psi^\dagger \psi \rangle = \mathrm{g}_\omega \rho_b.$$

$$m_{\boldsymbol{\rho}}^2 \langle \rho_3^0 \rangle = \mathrm{g}_{\boldsymbol{\rho}} \frac{(\rho_n - \rho_p)}{2}.$$

Final field equations

Effect of $U(\sigma)$ and $\boldsymbol{\rho}_\mu$ meson

- The field equations for σ, ω_μ and $\boldsymbol{\rho}_\mu$ fields are given by

$$m_\sigma^2 \langle \sigma \rangle = \mathrm{g}_\sigma \langle \overline{\psi}\psi \rangle = \mathrm{g}_\sigma \frac{2}{\pi^2} \int_0^{k_f} p^2 dp \frac{m - \mathrm{g}_\sigma \sigma}{\sqrt{p^2 + (m - \mathrm{g}_\sigma \sigma)^2}} - \mathrm{g}_2 \sigma^2 - \mathrm{g}_3 \sigma^3,$$

$$m_\omega^2 \langle \omega \rangle = \mathrm{g}_\omega \langle \psi^\dagger \psi \rangle = \mathrm{g}_\omega \rho_b.$$

$$m_{\boldsymbol{\rho}}^2 \langle \rho_3^0 \rangle = \mathrm{g}_{\boldsymbol{\rho}} \frac{(\rho_n - \rho_p)}{2}.$$

- The Dirac equation is modified as

$$\left[\gamma^\mu \left(p_\mu - \mathrm{g}_\omega \omega_\mu - \frac{1}{2} \mathrm{g}_{\boldsymbol{\rho}} \tau_3 \rho_{3,\mu} \right) - (m - \mathrm{g}_\sigma \sigma) \right] \psi = 0$$

Final field equations

Effect of $U(\sigma)$ and $\boldsymbol{\rho}_\mu$ meson

- The field equations for σ, ω_μ and $\boldsymbol{\rho}_\mu$ fields are given by

$$m_\sigma^2 \langle \sigma \rangle = \mathrm{g}_\sigma \langle \overline{\psi}\psi \rangle = \mathrm{g}_\sigma \frac{2}{\pi^2} \int_0^{k_f} p^2 dp \frac{m - \mathrm{g}_\sigma \sigma}{\sqrt{p^2 + (m - \mathrm{g}_\sigma \sigma)^2}} - \mathrm{g}_2 \sigma^2 - \mathrm{g}_3 \sigma^3,$$

$$m_\omega^2 \langle \omega \rangle = \mathrm{g}_\omega \langle \psi^\dagger \psi \rangle = \mathrm{g}_\omega \rho_b.$$

$$m_{\boldsymbol{\rho}}^2 \langle \rho_3^0 \rangle = \mathrm{g}_{\boldsymbol{\rho}} \frac{(\rho_n - \rho_p)}{2}.$$

- The Dirac equation is modified as

$$\left[\gamma^\mu \left(p_\mu - \mathrm{g}_\omega \omega_\mu - \frac{1}{2} \mathrm{g}_{\boldsymbol{\rho}} \tau_3 \rho_{3,\mu} \right) - (m - \mathrm{g}_\sigma \sigma) \right] \psi = 0$$

which extracts the eigen value as

$$e_{I_3}(p) = \mathrm{g}_\omega \omega_0 + \mathrm{g}_{\boldsymbol{\rho}} \rho_3^0 \frac{1}{2} (\overline{\psi} \gamma^0 \tau_3 \psi)_{\boldsymbol{p}_\kappa} + \sqrt{p^2 + (m - \mathrm{g}_\sigma \sigma)^2}$$

Energy density and pressure

$\sigma - \omega - \rho$ model

Energy density and pressure

$\sigma - \omega - \rho$ model

- So the energy density of the $\sigma - \omega - \boldsymbol{\rho}$ model is given by

$$\epsilon = \mathscr{H}_\sigma + \mathscr{H}_\omega + U(\sigma) + \mathscr{H}_{\boldsymbol{\rho}} + \mathscr{H}_{\psi+int}$$

Energy density and pressure

$\sigma - \omega - \rho$ model

- So the energy density of the $\sigma - \omega - \boldsymbol{\rho}$ model is given by

$$\begin{aligned}\epsilon &= \mathscr{H}_\sigma + \mathscr{H}_\omega + U(\sigma) + \mathscr{H}_{\boldsymbol{\rho}} + \mathscr{H}_{\psi+int} \\ &= \frac{1}{2}m_\sigma^2\sigma^2 - \frac{1}{2}m_\omega^2\omega_0^2 - \frac{1}{2}m_{\boldsymbol{\rho}}^2\rho_{0,3}^2 + \frac{1}{3}\mathrm{g}_2\sigma^3 + \frac{1}{4}\mathrm{g}_3\sigma^4 + \langle\overline{\psi}\gamma_0 p_0\psi\rangle\end{aligned}$$

Energy density and pressure

$\sigma - \omega - \rho$ model

- So the energy density of the $\sigma - \omega - \boldsymbol{\rho}$ model is given by

$$\begin{aligned}\epsilon &= \mathscr{H}_\sigma + \mathscr{H}_\omega + U(\sigma) + \mathscr{H}_{\boldsymbol{\rho}} + \mathscr{H}_{\psi+int} \\ &= \frac{1}{2}m_\sigma^2\sigma^2 - \frac{1}{2}m_\omega^2\omega_0^2 - \frac{1}{2}m_{\boldsymbol{\rho}}^2\rho_{0,3}^2 + \frac{1}{3}\mathrm{g}_2\sigma^3 + \frac{1}{4}\mathrm{g}_3\sigma^4 + \langle\overline{\psi}\gamma_0 p_0\psi\rangle\end{aligned}$$

- Now, we can write

$$\langle\overline{\psi}\gamma_0 p_0\psi\rangle = \frac{1}{\pi^2}\sum_{I_3}\int_0^{k_f} p^2\,dp\;e_{I_3}(p)\Theta[\mu - e_{I_3}(p)]$$

Energy density and pressure

$\sigma - \omega - \rho$ model

- So the energy density of the $\sigma - \omega - \boldsymbol{\rho}$ model is given by

$$\begin{aligned}\epsilon &= \mathscr{H}_\sigma + \mathscr{H}_\omega + U(\sigma) + \mathscr{H}_{\boldsymbol{\rho}} + \mathscr{H}_{\psi+int} \\ &= \frac{1}{2}m_\sigma^2\sigma^2 - \frac{1}{2}m_\omega^2\omega_0^2 - \frac{1}{2}m_{\boldsymbol{\rho}}^2\rho_{0,3}^2 + \frac{1}{3}\mathrm{g}_2\sigma^3 + \frac{1}{4}\mathrm{g}_3\sigma^4 + \langle\overline{\psi}\gamma_0 p_0\psi\rangle\end{aligned}$$

- Now, we can write

$$\langle\overline{\psi}\gamma_0 p_0\psi\rangle = \frac{1}{\pi^2}\sum_{I_3}\int_0^{k_f} p^2\,dp\; e_{I_3}(p)\Theta[\mu - e_{I_3}(p)]$$

$$\begin{aligned}= &\frac{1}{\pi^2}\int_0^{k_n} p^2\,dp\left[\mathrm{g}_\omega\omega_0 + \frac{1}{2}\mathrm{g}_{\boldsymbol{\rho}}\rho_3^0 + \sqrt{p^2 + (m - \mathrm{g}_\sigma\sigma)^2}\right] \\ &+\frac{1}{\pi^2}\int_0^{k_p} p^2\,dp\left[\mathrm{g}_\omega\omega_0 - \frac{1}{2}\mathrm{g}_{\boldsymbol{\rho}}\rho_3^0 + \sqrt{p^2 + (m - \mathrm{g}_\sigma\sigma)^2}\right]\end{aligned}$$

Energy density and pressure

$\sigma - \omega - \rho$ model

- So the energy density of the $\sigma - \omega - \boldsymbol{\rho}$ model is given by

$$\begin{aligned}\epsilon &= \mathscr{H}_\sigma + \mathscr{H}_\omega + U(\sigma) + \mathscr{H}_{\boldsymbol{\rho}} + \mathscr{H}_{\psi+int} \\ &= \frac{1}{2}m_\sigma^2\sigma^2 - \frac{1}{2}m_\omega^2\omega_0^2 - \frac{1}{2}m_{\boldsymbol{\rho}}^2\rho_{0,3}^2 + \frac{1}{3}\mathrm{g}_2\sigma^3 + \frac{1}{4}\mathrm{g}_3\sigma^4 + \langle\overline{\psi}\gamma_0 p_0\psi\rangle\end{aligned}$$

- Now, we can write

$$\langle\overline{\psi}\gamma_0 p_0\psi\rangle = \frac{1}{\pi^2}\sum_{I_3}\int_0^{k_f} p^2\,dp\; e_{I_3}(p)\Theta[\mu - e_{I_3}(p)]$$

$$\begin{aligned}&= \frac{1}{\pi^2}\int_0^{k_n} p^2\,dp\left[\mathrm{g}_\omega\omega_0 + \frac{1}{2}\mathrm{g}_{\boldsymbol{\rho}}\rho_3^0 + \sqrt{p^2 + (m - \mathrm{g}_\sigma\sigma)^2}\right] \\ &\quad + \frac{1}{\pi^2}\int_0^{k_p} p^2\,dp\left[\mathrm{g}_\omega\omega_0 - \frac{1}{2}\mathrm{g}_{\boldsymbol{\rho}}\rho_3^0 + \sqrt{p^2 + (m - \mathrm{g}_\sigma\sigma)^2}\right] \\ &= m_\omega^2\omega_0^2 + m_{\boldsymbol{\rho}}^2\rho_{0,3}^2 + \frac{1}{\pi^2}\left[\int_0^{k_n} p^2\,dp\sqrt{p^2 + m^{*2}} + \int_0^{k_p} p^2\,dp\sqrt{p^2 + m^{*2}}\right]\end{aligned}$$

Energy density and pressure

$\sigma - \omega - \rho$ model

Energy density and pressure

$\sigma - \omega - \rho$ model

- So the energy density of the $\sigma - \omega - \boldsymbol{\rho}$ model is combined as

$$\begin{aligned}\epsilon &= \frac{1}{2}m_\sigma^2\sigma^2 + \frac{1}{3}g_2\sigma^3 + \frac{1}{4}g_3\sigma^4 + \frac{1}{2}m_\omega^2\omega_0^2 + \frac{1}{2}m_{\boldsymbol{\rho}}^2\rho_{0,3}^2 \\ &+ \frac{1}{\pi^2}\left[\int_0^{k_n} p^2\,dp\sqrt{p^2+m^{*2}} + \int_0^{k_p} p^2\,dp\sqrt{p^2+m^{*2}}\right]\end{aligned}$$

Energy density and pressure

$\sigma - \omega - \rho$ model

- So the energy density of the $\sigma - \omega - \boldsymbol{\rho}$ model is combined as

$$\begin{aligned}\epsilon &= \frac{1}{2}m_\sigma^2\sigma^2 + \frac{1}{3}g_2\sigma^3 + \frac{1}{4}g_3\sigma^4 + \frac{1}{2}m_\omega^2\omega_0^2 + \frac{1}{2}m_{\boldsymbol{\rho}}^2\rho_{0,3}^2 \\ &+ \frac{1}{\pi^2}\left[\int_0^{k_n} p^2\,dp\sqrt{p^2+m^{*2}} + \int_0^{k_p} p^2\,dp\sqrt{p^2+m^{*2}}\right]\end{aligned}$$

- In same fashion $\mathtt{p}$ of the $\sigma - \omega - \boldsymbol{\rho}$ model is written as

$$\begin{aligned}\mathtt{p} &= -\frac{1}{2}m_\sigma^2\sigma^2 - \frac{1}{3}g_2\sigma^3 - \frac{1}{4}g_3\sigma^4 + \frac{1}{2}m_\omega^2\omega_0^2 + \frac{1}{2}m_{\boldsymbol{\rho}}^2\rho_{0,3}^2 \\ &+ \frac{1}{3\pi^2}\left[\int_0^{k_n} p^2\,dp\frac{dp^2}{\sqrt{p^2+m^{*2}}} + \int_0^{k_p} p^2\,dp\frac{dp^2}{\sqrt{p^2+m^{*2}}}\right]\end{aligned}$$

References:

- Compact Stars, Springer, **Norman K. Glendenning**.
- Quantum Field Theory, Cambridge University Press, **Lewis H. Ryder**.